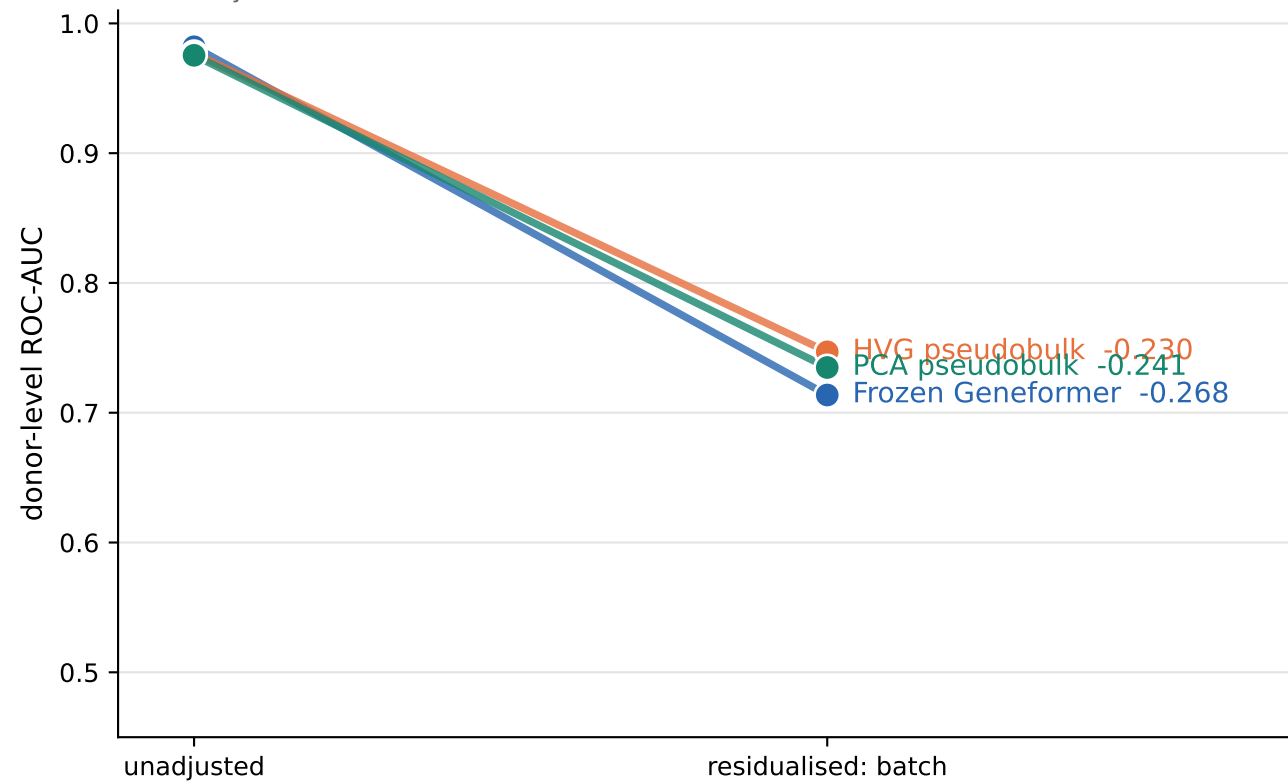


Figure 4 Residualisation is a sensitivity operation, not a causal batch-effect estimate

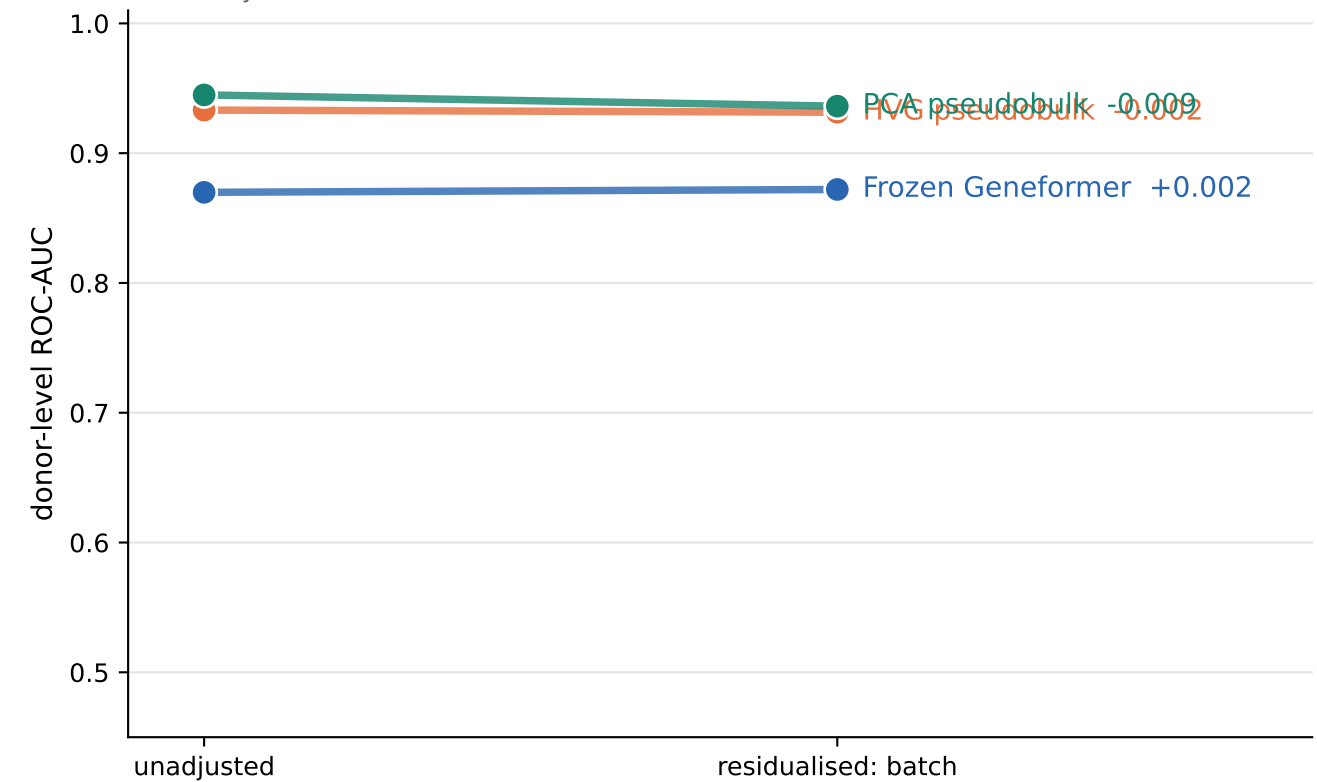
A Batch residualisation removes 0.23-0.27 AUC

batch-only disease AUC = 0.922



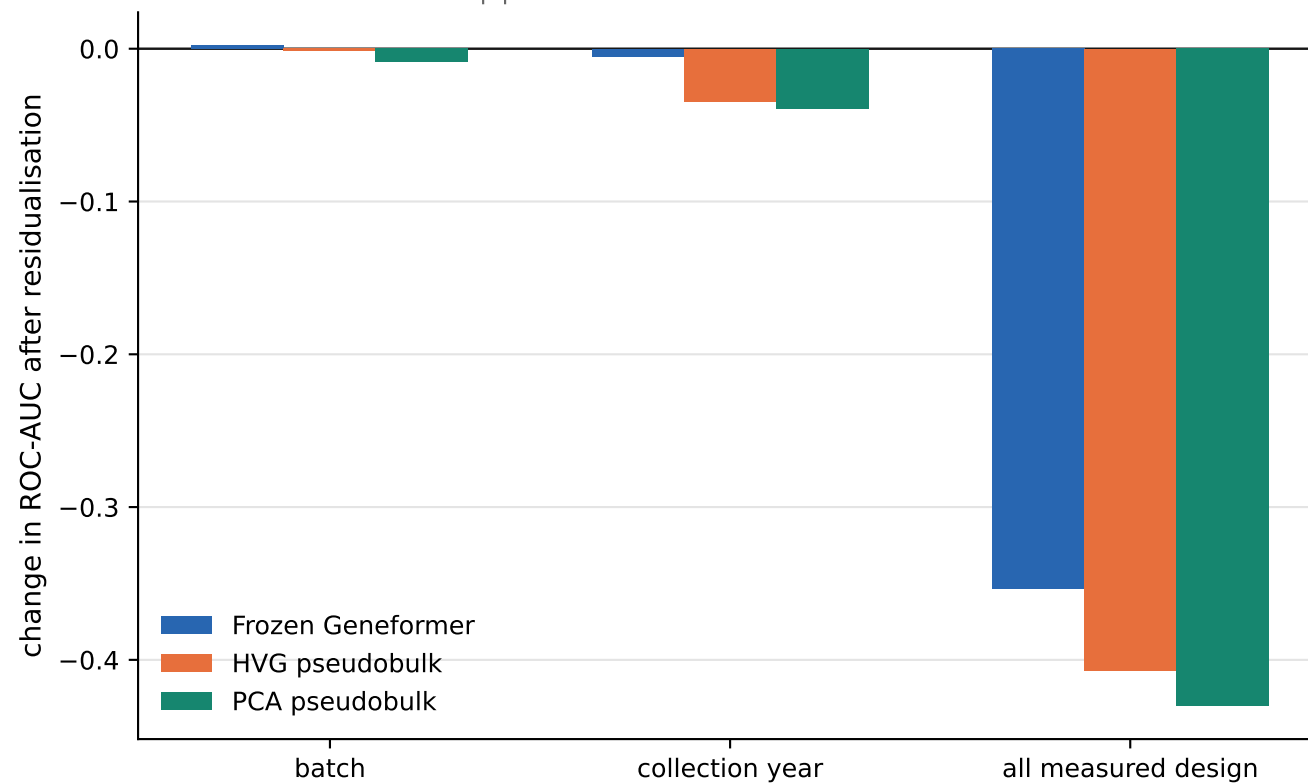
B Batch residualisation changes almost nothing

batch-only disease AUC = 0.499



C The answer depends on which design block is projected out

identical folds and classifier pipeline



D Design predictability forecasts residualisation damage

descriptive relationship across prespecified blocks

